Electronic-Photonic Co-Design of Co-Packaged Optics in Schematic-Driven Layout Design Flow for Photonic Integrated Circuits

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***Abstract*—The transition from pluggable optics to co-packaged optics is having an impact on the photonic integrated circuit design automation. Traditionally, chip designers for fiber-optic systems using electronic design automation tools frequently needed a way to model a few photonic components in the same design environment with the electronics. Examples include modeling lasers or modulators with electrical driver circuits, or modeling receiver electronics with photodetector. These designs focused primarily on optimizing electronics and it was typically sufficient to model a few photonic components through electrically equivalent circuit representations using SPICE or Verilog-A. However, with the advances in silicon photonics, the photonic component count in modern PICs is rapidly growing raising serious concerns over efficacy and reliability of treating photonics as electronics. This paper describes an E-O co-design approach that (i) does not require user intervention on RF and photonic domain demarcations, and (ii) supports schematic-driven layout and back-annotation.**

Keywords—photonic integrated circuits, co-packaged optics, silicon photonics, E-O co-design, schematic-driven layout, process design kits

I. Introduction

The application space for photonic integrated circuits (PICs) is rapidly expanding. Some of the civilian, aerospace and defense interests in PICs range from traditional telecom transceivers to automotive, sensing, consumer wearables, high-performance computing (HPC), artificial intelligence (AI), and quantum information science (QIS). In addition to the favorable economics from higher spectral and energy efficiencies, smaller footprint and lower cost, photonic integration and manufacturing readiness are recognized as essential to the US national security by the Department of Defense (DoD) [1].

Silicon Photonics (SiPh) facilitates transfer the knowledge gained from decades of manufacturing experience in electronic integrated circuits to photonic integration. SiPh is also a key driver to bringing electronics closer to the photonics in the same package. The push for the co-packaged optics (CPO) and explosive growth of the component count and complexities in today's PICs require rethinking of the electro-optic (E-O) co-design practices. Traditionally, chip designers for fiber-optic systems using electronic design automation (EDA) tools frequently needed a way to model a few photonic components in the same design environment with the electronics. Examples include modeling lasers or modulators with electrical driver circuits, or modeling receiver electronics with photodetector. These designs focused primarily on optimizing electronics and it was typically sufficient to model a few photonic components through electrically equivalent circuit representations using SPICE or Verilog-A [2]. As the interest in PICs picked up momentum, early electro-optical co-design approaches continued [3-6] since the component count and complexities remained relatively low. As we shall see in the next sections of this paper, it is no longer enough to model photonic components through electrically equivalent circuit representations in SPICE or Verilog-A.

The paper is organized as follows. Section II discusses fundamental differences between the optical and the electrical signals. In the following section, we describe an equivalent electrical representation of a photonic model. Section IV of the paper focuses on the trade-offs and repercussions of treating photonics as electronics. We make a case for using domain-specific electrical and photonic circuit simulators in Section V. An example is used to illustrate the design simulation flow using commercially available electrical and photonic circuit simulators.

II. Optical vs. Electrical Signals

Electrical signals describe current and voltage in the RF domain and are real-valued, baseband signals. A baseband signal has non-zero spectral contents near DC and is lowpass. If $V(t)$ represents an electrical signal, its Fourier transform is Hermitian and can be described by the positive frequency part of the spectrum.

$$V(f) = V^{*}(-f) \qquad (1)$$

Optical signals, on the other hand, are conveniently represented as analytic, narrowband signals with a complex-valued envelope, i.e., a complex number with real and imaginary parts describing amplitude and phase modulation of the optical carrier. An optical signal is typically modeled with the complex envelope and its center frequency as two distinct pieces of information.

$$E(t) = A(t)e^{-j2\pi f_0 t} + A^{*}(t)e^{j2\pi f_0 t} \qquad (2)$$

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where E(t) is the amplitude function of the optical signal, f_0 is the center frequency, and A(t) is the envelope function.

The optical carrier can be a single-wavelength or a broadband, multi-wavelength signal propagating with associated polarization states and transverse mode profiles.

For the vast majority of optical components in communications systems and photonic circuits, it is acceptable to work with only the transverse components of the field $[E_x(t), E_y(t)]$ -- the longitudinal component $E_z(t)$ is typically negligible and may be recovered from the transverse components anyway. Thus the envelope representation for the field is extended to the vector form:

$$\boldsymbol{E(t)} = \gamma f(x,y)\{\hat{x}[A_x(t)e^{-2\pi j f_0 t} + A_x(t)^* e^{2\pi j f_0 t}] + \hat{y}[A_y(t)e^{-2\pi j f_0 t} + A_y(t)^* e^{2\pi j f_0 t}]\} \quad (3)$$

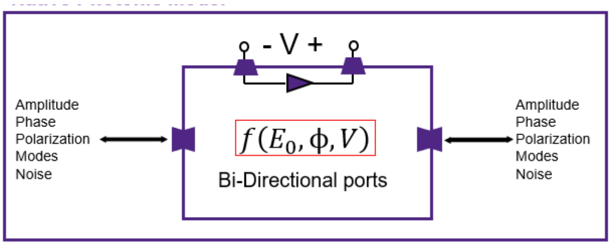
where $f(x,y)$ represents spatial mode profile. The scaling factor γ in Eq. (3) allows us to choose convenient units for the amplitude function. It is conventional to define γ such that the squared amplitude $|A(t)|^2$ has units of Watts. The vector envelope quantity can be represented as

$$A(t) = \begin{bmatrix} A_x(t) \\ A_y(t) \end{bmatrix} \quad (4)$$

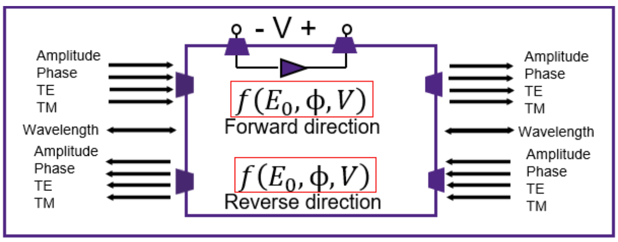
The two-component column in Eq. (4) is referred to as the "Jones vector" representation.

III. Representing a Photonic Model in an Equivalent Electrical Formalism

In the preceding section, we saw that the electrical and optical signals are fundamentally different. An optical signal is very rich in its attributes. In order to represent a photonic model with bidirectional, multiwavelength, multimode optical signal in an equivalent electrical analogy requires treating each of the optical signal attributes as individual electrical signals and pins, a concept widely referred to as "port expansion." Figure 1 illustrates the mapping.



(a) A native, photonic model



(b) An equivalent electrical representation of (a)

Fig. 1. Representing an optical signal in electrical domain requires up to a nine-fold increase per wavelength in the number of electrical ports and signals for every photonic device in the circuit

Electrical circuit simulators for analog electronics are based on modified nodal analysis and Kirchhoff's electrical circuit laws [7-8]. Photonic circuits create closed propagation paths either due to the circuit itself (e.g., coupled ring resonators and modulators) or through multiple orders of reflections and multipath interference (MPI) from neighboring components. The vectoral nature of the optical field waves results in coherent interactions between signals. Modeling these effects in the electrical domain using Kirchhoff's voltage and current (both scalar quantities) conservation laws results in unintuitive, resource-consuming complexities to the development and maintenance of foundry process design kit (PDK), and subsequently to the design of PICs.

IV. Consequences of Electrical Representation of Photonics

The fact that a photonic circuit can have a large number of components, operating at multiple wavelengths, each wavelength with multiple transverse modes per polarization and direction of propagation makes equivalent electrical representation via port expansion of Fig. 1(b) exorbitantly expensive and potentially error-prone.

The electrical representation of optical signals also poses challenges for probing optical quantities at the points of interest in a circuit. For example, measurement and visualization of optical quantities such as polarization mode dispersion, polarization dependent losses, interplay between optical noise, dispersion, nonlinearity and polarization – all of which are also wavelength and polarization dependent, inter- and intra-channel crosstalk, mode-coupling, etc. come at the expense of extensive, error-prone post-processing on probed electrical signals.

Many applications require creation and use of customized photonic components either for augmenting a foundry's interoperable PDK (iPDK), or for creating an IP. It's common practice in the industry to use applicable optical mode-solvers during photonic device design. Forcing photonic device simulators to produce an analogous electrical cell can severely hamper the scope and opportunity of creating and using custom photonics.

In a schematic-driven layout (SDL) design environment, even when multiple electrical signals (mapping various properties of the optical signal) can be collectively represented as buses, multiwavelength and multimode photonic devices can dramatically increase the signal and pin count. This creates serious challenges for tracking, connecting (crossings, for example), and probing signals at the pin and bus levels. In addition, the likelihood of layout-vs-schematic check (LVS) violations dramatically increases causing verification delays.

V. Co-Design in Native E-O Signal Domains

The main motive for representing a photonic component in electrical domain is to be able to use a single electrical circuit simulator for both electronics and photonics on the same chip. This early practice emanated from the lack of mature photonic circuit simulators in past. However, the domain-specific simulators for electronic and photonic circuits are mature today and are available from multiple vendors. Using an electrical simulator for the electrical sub-circuit of a PIC, and a photonic simulator for the photonic sub-circuit of the PIC eliminates expensive trade-offs one would require when solving the entire PIC in electrical domain. The widely adopted slowly varying envelope representation of optical signals eliminates the need for terahertz rate sampling of electrical signals. Therefore, both optical and electrical signals can be sampled on a common grid at the modulation speeds - in the same simulation session - without additional computational overheads.

A. Example

As an illustration of the E-O co-design using native E-O circuit simulators, consider a traveling-wave Mach-Zehnder modulator (TW-MZM) PIC of Fig. 2.

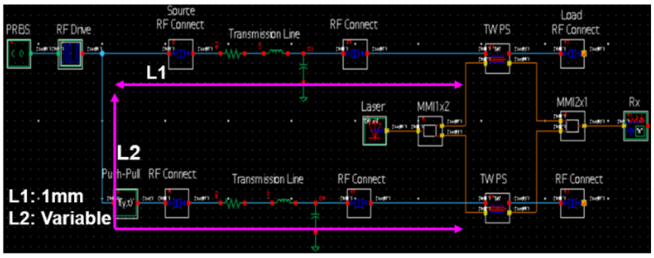


Fig. 2. Circuit schematic of a TW-MZM to study impact of RF path-length differences due to asymmetrical locations of electrical pads and phase-shifters

The schematic of Fig. 2 contains both photonic (laser, splitter, coupler, travelling-wave phase-shifters) and electronic (signal generators, source and load to the electrodes, and electrical connections) circuit elements. Since electronics and photonics are tightly coupled in travelling-wave fashion, any path-length deviation between the phase-shifter electrodes and electrical pads for connecting RF source and load would impact the bandwidth of the TW-MZM PIC. Such path length differences can arise from packaging and wafer area constraints, analyses of which is extremely important to layout engineers in order to avoid or minimize unwarranted performance penalties.

B. Using Domain-specific Circuit Simulators

Since the circuit of Fig. 2 contains both electronic and photonic PDK elements, we use Synopsys HSPICE for simulating the electrical sub-circuit and Synopsys OptSim for simulating photonic sub-circuit. The subcircuits are identified and partitioned automatically by the Synopsys OptoCompiler design environment.

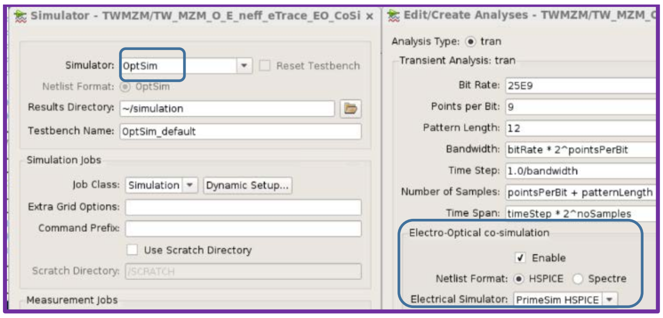


Fig. 3. Specifying photonic and electronic circuit simulators for respective subcircuits

The simulation setup (Fig. 3) also includes a common sampling grid, design variables and probes. Both electrical and photonic waveforms can be probed and viewed in the same simulation run.

Fig. 4 shows a plot of back-to-back receiver Q-factor versus the mismatch in electrical path lengths connecting each electrode to the modulating RF signal source.

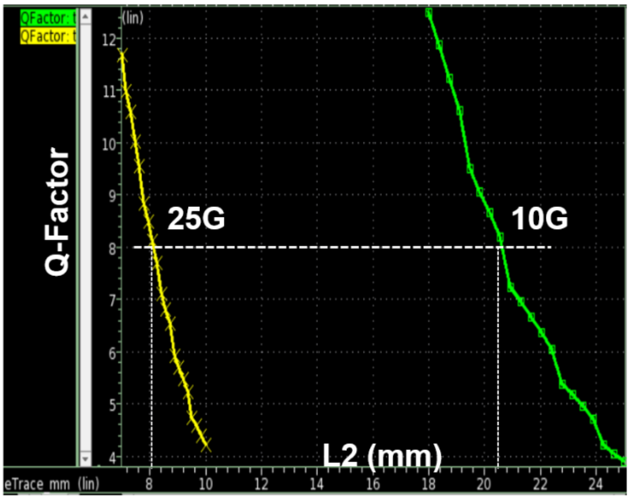


Fig. 4. Q-factor vs. length of one of the electrical traces connecting an RF source to the electrode

As evident from the plot, in order to guarantee a Q-factor of 8, an RF path difference as large as 21mm is acceptable at modulation speed of 10Gbps. As expected, faster modulation speeds significantly tighten the acceptable tolerances in path-length deviations and only an 8mm of deviation is acceptable at 25Gbps for the PIC under study.

VI. Conclusion

Rapidly increasing component count in modern PICs and the push for CPO make treating photonics as electronics a risky approach for electronic-photonic co-design. Optical signals are complex and rich in attributes quite unlike the electrical signals. Superfluous means to attach optical signal properties to an electrical signal lead to explosive pin count creating unnecessary design complexities and overheads in modeling, probing, and verification. Availability of mature, domain-specific, electrical, and photonic design tools from multiple vendors today makes a compelling case for using domain-specific circuit simulators making the overall E-O design more seamless, scalable, tractable, intuitive and faithful to the physics of the circuit.

References

[1] C. Pellerin, “DoD competition to launch new manufacturing institute,” DoD News article, https://www.defense.gov/News/News-Stories/Article/Article/603407/ Oct. 2014

[2] E. Kononov, “Modeling photonic links in Verilog-A,” Master’s Thesis, MIT, June 2013.

[3] P. Martin, F. Gays, E. Grellier, and S. Menezo, “Development of Verilog-A models for silicon photonics devices and implementation in a standard EDA environment,” MOS-AK Workshop, Présentation PowerPoint (mos-ak.org), March 28-29, 2014.

[4] C. Sorace-Agaskar, J. Leu, M. Watts, and V. Stonanovic, “Electro-optical co-simulation for integrated CMOS photonic circuits with VerilogA,” Optics Express, vol.23, no. 21, Oct. 2015.

[5] S. Lin, “Electronic-photonic co-design of silicon photonic interconnects,” Ph D Thesis, UC-Berkeley, Fall 2017.

[6] J. Leu, “Integrated silicon photonic circuit simulation,” Ph D Thesis, MIT, Sept. 2018.

[7] C. W. Ho, A. Ruehli, and P. Brennan, “The modified nodal approach to network analysis,” IEEE Trans. on Circuits and Systems, vol. CAS-22, no. 6, pp. 504-509, June 1975.

[8] M. Fiers, T. V. Vaerenbergh, K. Caluwaerts, D. V. Ginste, B. Schrauwen, J. Dambre, and P. Bienstman, “Time-domain and frequency-domain modeling of nonlinear optical components at the circuit-level using a node-based approach,” J. of OSA B, vol. 29, no. 5, pp. 896-900, May 2012.